Recursion

1st year → 3rd year
→ tell your story
1 month

→ most important topic.

→ how not to give up.

→ help us to solve complex problem in a very easy way.

→ will be a lot useful in future for lots of data structure and DP.

→ how not to give up on recursion.

We might face some difficulties in starting but once it becomes clear, recursion is very easy.

first, let us quickly revise concept of function → why fn
→ DRY principle
→ python tutor

Can a function call itself as well? → Yes
→ Stack memory
→ Concept of stack overflow

Now with above exercise, 2 things are clear

① Functions wait in memory unless they are resolved

② When a fn finishes execution, then only it comes out of program and gets deleted from stack.

Recursion is function calling itself.

fun1() → fun2()

fun1() → fun1() → recursion

"Recursion is when solution of a problem depends on some smaller program"

factorial of a no.

$$n! = n * (n-1) * (n-2) * (n-3) * \dots 1$$

$$= n * (n-1)!$$

$$\boxed{fact(n) = n * fact(n-1)}$$

explain here how this is a recursion problem & relation

Now, $fact(n-1) = (n-1) * (n-2) * (n-3) * \dots 1$

$$= (n-1) * fact(n-2)$$

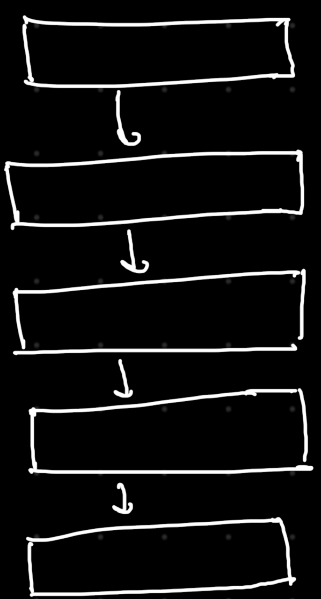
We faced a recursion limit exceeded error } explain this & along with C++/Java

line no.

Show on net as well

fact(5)
↓
fact(4)
↓
fact(3)
↓
fact(2)
↓
fact(1)
↓
fact(0) → fact(-1) → fact(-2) → fact(-3) . . . .

prove this using print(n)



Each and every fn is an instance & has diff value of n

So we have to make sure that recursion stops somewhere and don't keep on going on till ∞.

↳ Let us add a base condn.

Now, we will not be making this fn calls and everything and think of recursion this deep.

let us understand about the underlying principle of recursion.

PMI (Principle of mathematical Induction)

Initial few problem we will do can be done iteratively but we have to use recursion to get mastery. and as it is way to easy like this.

Recursion principle based on PMI, so understanding this will further help us to master recursion.

Show book

PMI

It is used to prove some fact.

eg.

$$\Sigma n = \frac{n(n+1)}{2} = sum(n)$$

how they are imp for recursion as well.

using PMI, we can break this problem in 3 steps.

1. Prove F(0) or F(1) is true.
2. Assume F(K) is true → Induction hypothesis
3. Using step 2, i.e. F(K) is true* prove that F(K+1) is true.

$$sum(n) = \Sigma n = \frac{n(n+1)}{2}$$

Let us prove above using PMI

① Base case F(0)

L.H.S. $\Sigma 0 = 0$

R.H.S. $0\frac{(0+1)}{2} = 0$

L.H.S = R.H.S

② Assume (don't Question) that F(K) is true

$\Rightarrow \Sigma R = sum(R) = \frac{R(R+1)}{2}$ ⇒ This is true

③ Now, we need to prove that (R+1) is true

$\Rightarrow \Sigma R+1 = sum(R+1) = \frac{(R+1)(R+2)}{2}$

L.H.S. $= \Sigma R+1 = R+1 + \Sigma R$

$= (R+1) + \frac{(R+1)R}{2}$

$= \frac{2*(R+1)}{2} + \frac{(R+1)R}{2}$

$= \frac{(R+1)(R+2)}{2}$ = R.H.S

Hence by using principle of mathematical induction we have proven that

$$\sum n = \frac{n(n+1)}{2}$$

Now, let us deep dive and see how exactly it works?



→ we also know how to go from a step to another step.



→ we know the base

So basically we know the base and how to take a step as well so we can move forward up the steps.



first explain visually and then via examples



So, $f(0) \rightarrow$ true

Say $k \rightarrow 0$ $\quad f(0) \rightarrow$ true

$\Rightarrow 1$ $\quad f(1) \rightarrow$ true

$f(2) \rightarrow$ true

So in every recursion problem we just have to apply the PMI and our work is done.

Just below 3 steps.

1. Base case
2. Assume for $(n-1)$ ~> small problem is true
3. Using assumption in step(2), solve for bigger problem.

if all the above steps are properly followed.

P.M.I. says that we will get the answer.

ideally, no need to make f^n calls again and again.

Now, we will solve lots of question to make it clear.

Q-1 Find 2 to the power of n for a given n, using recursion tree.

→ Also state recursive function and make recursion tree.

again it can be solved via iteration but we are going to use recursion.

Q-2 Find no. of digits in a given number.

234 → 3

51234 → 5

draw recursion tree and recursion formula.

Q-3.

Fibonicci Series